

Hanoi Urban Noise Mapping

Data Collection Report

363 smartphone measurements · 3 sites · 2026-06-10 to 2026-07-22

Measurements	Sites	$L_{A,25s}$ range	Model R^2	> QCVN day
363	3	47–88	0.246	37.4%

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1 Objective

Map and characterise urban noise in three contrasting districts of Hanoi, and predict noise from urban morphology. We follow the Sunbird AI methodology (*Urban Noise Uganda 61K*, Nsumba et al., 2026): smartphone field collection plus a morphology-to-noise model, reproduced first on their data and then applied here.

Measurement status — read this before any figure below

*Our target quantity is a 20–30 s A-weighted level from consumer smartphones, denoted $L_{A,25s}$. It is **not** a certified L_{Aeq} , and not L_{den} or L_{night} . The three phones were cross-calibrated against each other, never against a reference instrument. Contrasts between places and between hours are supported; absolute levels are indicative, with a plausible bias of +3.0 to +7.3 dB (section 3.4). No compliance claim is made anywhere in this report.*

2 Data collection summary

Site	n	Median dB	Min–Max	% < 60 dB	% roadside
Hoan Kiem lake	99	70	55–80	7%	89%
Ocean Park	184	68	47–88	28%	73%
Vinh Tuy area	80	66	53–76	20%	74%
All	363	68	47–88	21%	78%

Table 1: Field campaign by site. *Roadside* pools the three survey bins that mean “within 10 m of a road”; the form changed mid-campaign and v1 offered a single 0–10 m bin where v2 splits 0–2 m from 2–10 m.

Zones. *Ocean Park* — new high-rise development, with shielding and active construction. *Vinh Tuy* — transport infrastructure, heavy traffic. *Hoan Kiem* — historic old quarter, narrow streets, pedestrian zones.

Quality control. Site labels were cross-checked against the GPS fix and six mislabelled submissions were reassigned to the nearest site centre. No row was dropped: the raw export holds 363 submissions and this dataset holds 363 rows.

Headline findings. Median 68 dB; 37.4% of short daytime samples sit above the QCVN daytime threshold value; levels peak at 18:00. The delivered model reaches $R^2 = 0.246$ [0.097, 0.339] under Buffered LOO 300 m (section 5), against baselines reported in the same table.

3 Methodology

3.1 Field data-collection protocol

Study zones were mapped in Google Earth and chosen for contrasting typologies (section 2). At each point we recorded a 20–30 s reading, with a 10 s audio clip, in ODK Collect, submitted live to a KoboToolbox server with an automatic timestamp and GPS fix. Phones were held at 1.2 m (SPB positioning) at varied distances from the road, by three cross-calibrated collectors, from 05:00 to 23:00 with emphasis on rush hours. Weather (Open-Meteo) and spatial context (OpenStreetMap) are joined afterwards.

3.2 Reproduction of the Sunbird pipeline

We reproduced the full Sunbird chain on their Uganda dataset (notebooks 01–06): cleaning, audio QC, morphology features, figures, surrogate model. Our statistics match the paper —

median 49 dB against their 45–50 dB.

3.3 Model features

Morphology within 300 m of each point, from OpenStreetMap: built-area ratio, road density, intersection count, and distance to the nearest road; plus hour of day and a weekend flag. The v2 feature set splits distance by road class (`dist_highway`, `dist_residential`) and encodes the hour cyclically.

3.4 Reference values and measurement status

Reference value	Day (6–21h)	Night (21–6h)
QCVN 26:2010/BTNMT (Vietnam, ordinary area)	70 dB	55 dB

The WHO road-traffic guideline values (53 / 45 dB) have been **withdrawn** from this report. They are L_{den} and L_{night} : annual averages carrying +5 dB evening and +10 dB night penalties. Comparing them to a 25 s daytime sample compares different statistics of different processes. QCVN is retained because it regulates a level — but it regulates an L_{Aeq} measured with a class 1–2 meter under TCVN 7878-2:2010, a specification our instrument does not meet either.

Consequence for every exceedance figure in this report

*It is a **descriptive statistic of our sample** — the share of short samples above a threshold value — and never a finding of regulatory non-compliance.*

Anchoring the absolute bias. Because we cannot calibrate against a reference instrument, we bound the bias by comparing our roadside-day stratum to instrumented campaigns, correcting for the difference expected from the quantity alone.

Source	Instrument	Quantity	Raw gap	Corrected
Gelb & Apparicio 2019, Applied Acoustics 148:332-343	personal dosimeters + GPS	$L_{\text{Aeq,1min}}$	+9.3	+7.3
Phan et al. 2010, Applied Acoustics 71(5):479-485	RION NL-21 / NL-22 (sound level meters), 24 h continuous	L_{den}	+7.0	+3.0
Phan et al. 2010, Applied Acoustics 71(5):479-485	RION NL-21 / NL-22, 24 h continuous	L_{night} (lowest of the 7 sites)	-3.0	-3.0
Institute of Occupational Health and Environment, 12 major Hanoi arteries (as reported by the Vietnamese press)	not documented	daytime mean dBA	+8.6	+7.6

Table 2: Anchoring on instrumented literature. The greyed row is grey literature — a secondary source with an undocumented protocol — and is excluded from the interval. Plausible absolute bias of our smartphones: **+3.0 to +7.3 dB**. It is reported and **never applied**: the calibration offset is 0.0 for all three collectors.

4 Data analysis

4.1 Temporal patterns

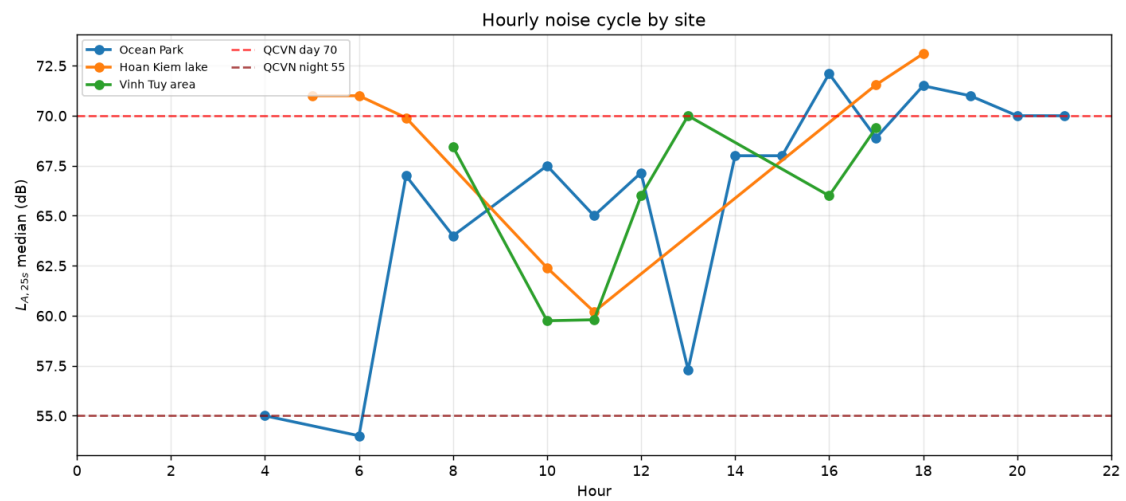


Figure 1: Hourly cycle of the median $L_{A,25s}$ by site. QCVN day and night reference values are drawn for orientation only.

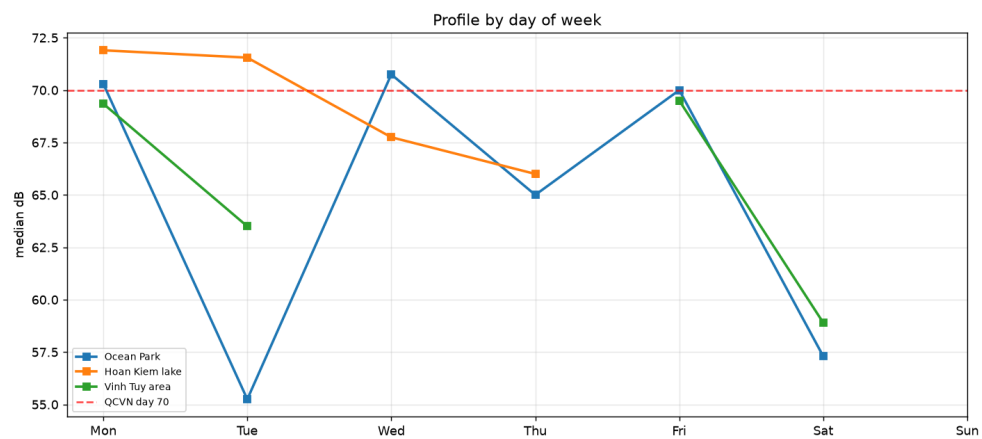


Figure 2: Day-of-week pattern. Weekend coverage is thinner than weekday coverage, and the difference should be read as indicative.

4.2 Sources and exceedances

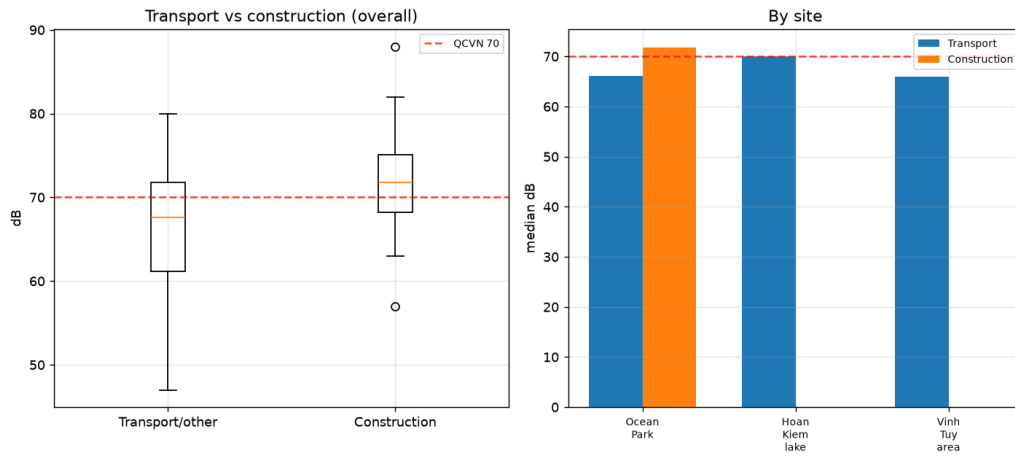


Figure 3: Level by declared noise category.

Site	Period	n	Median dB	% above threshold	Mean excess dB
Hoan Kiem lake	day	93	70.1	50.5%	3.9
Hoan Kiem lake	night	6	71.0	100.0%	15.2
Ocean Park	day	180	68.0	35.6%	4.4
Ocean Park	night	4	57.0	50.0%	9.5
Vinh Tuy area	day	80	65.9	26.2%	1.9

Table 3: Share of short samples above the QCVN threshold value, by site and period. Night rows rest on very few points (10 measurements fall between 21:00 and 06:00 in the whole campaign) and are reported for completeness, not for inference.

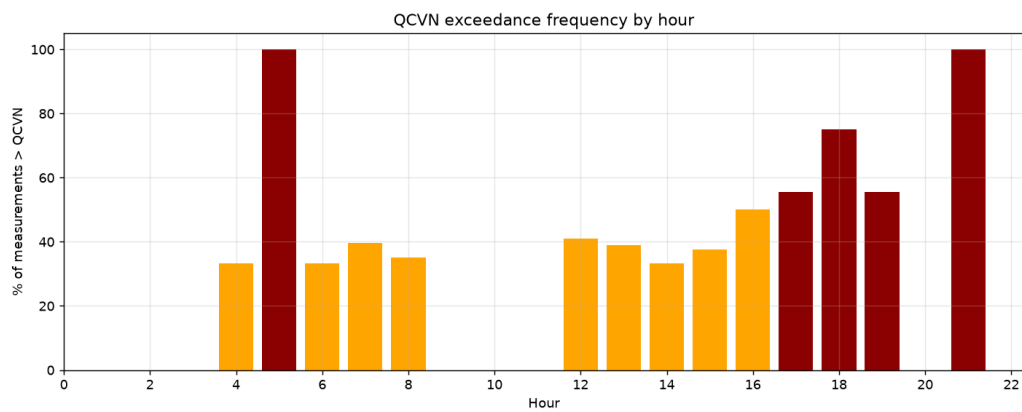


Figure 4: Share of short samples above the QCVN daytime threshold value, by hour. A bias of a few decibels moves this curve substantially, which is why the interval of section 3.4 is quoted with it.

4.3 Traffic and weather

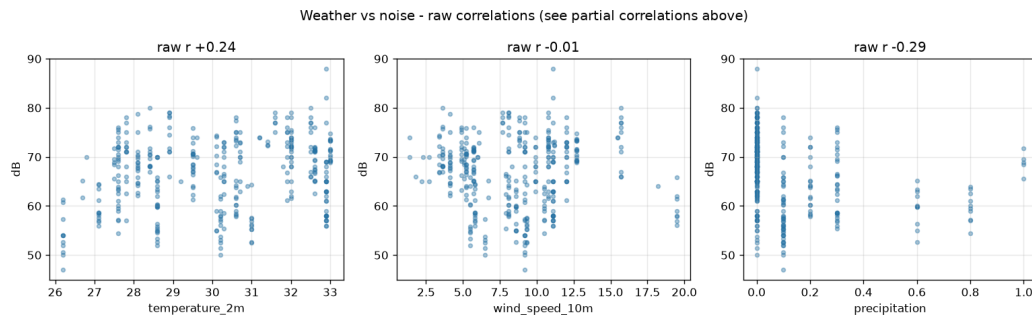


Figure 5: Weather covariates against level. Temperature, wind and precipitation are joined from Open-Meteo at the hour of measurement.

Traffic videos. 147 timestamped videos were counted with YOLOv8 and ByteTrack, producing flow by class from line crossings rather than density. Non-negative energy regression against the matched levels returns **zero coefficients** for motorcycles and cars. Two causes are identifiable and neither is acoustic: speed is unobservable without a ground homography, and the field of view is not georeferenced, so the source–receiver distance is discarded.

The detector has never been validated against manual counts

*Precision, recall and MAPE per class are unknown, and two-wheeler under-detection is expected but unquantified. **Modal shares must not be published until it is done.** One day of manual counting on ten videos turns an open-ended limitation into a quantified uncertainty; it is the highest value-per-effort task left in the project.*

5 Predictive model

5.1 How performance is measured

An earlier version of this report quoted $R^2 = 0.45$ from a cross-validation grouped on cells of about 110 m. The morphology features aggregate over a disc of **radius 300 m**: two points 110 m apart share more than 85 % of that disc, so the model saw near-twins of its test points during training. That score was not out-of-sample and **has been withdrawn** (Roberts et al. 2017, *Ecography*).

Every model below is scored under three splits, on exactly the same folds:

1. **Block-CV 600 m** — square blocks in UTM 48N, `GroupKFold` on the block identifier. 17 blocks. The permissive protocol.
2. **Buffered leave-one-out 300 m** — for each point, train on all points more than 300 m away from it. **The reference protocol**: it matches the feature radius and has no split randomness.
3. **Leave-one-site-out** — generalisation to an unseen urban typology.

Every score carries a 95 % interval bootstrapped **by block** over 2000 resamples, not by point: resampling correlated points would give an interval far too narrow.

5.2 Model comparison

Model	Block-CV	Buffered LOO	95 % CI	LOSO	MAE
	R^2	R^2 (<i>reference</i>)		R^2	dB
global_mean	-0.019	-0.039	[-0.118, -0.029]	-0.058	6.06
site_mean	-0.027	-0.198	[-0.546, -0.030]	-0.058	6.62
site_hour_mean	-0.008	-0.419	[-0.772, +0.031]	-0.058	6.58
dist_road	+0.221	+0.200	[+0.077, +0.266]	+0.189	5.28
idw	+0.117	-0.203	[-0.371, +0.022]	-0.140	6.44
lgbm_time	+0.230	+0.075	[-0.131, +0.460]	-0.139	5.19
lgbm_morpho	+0.153	+0.041	[-0.243, +0.231]	+0.007	5.63
lgbm_full	+0.304	+0.137	[-0.101, +0.332]	+0.029	5.26
lgbm_v2	+0.332	+0.099	[-0.130, +0.250]	-0.035	5.54
physical	+0.255	+0.246	[+0.097, +0.339]	+0.222	5.01
hybrid	+0.395	+0.123	[-0.219, +0.295]	+0.035	5.39
hybrid_lowcap	+0.378	+0.144	[-0.148, +0.346]	+0.106	5.30

Table 4: All models on identical splits. The delivered model is in bold and is selected **by code** under the reference protocol; the choice is written into `models/metrics.json` and read by the export, so nothing downstream can silently substitute another.

The ranking inverts between the permissive split and the strict ones. The hybrid architecture — a physical core corrected by a LightGBM fitted to the residual, the design this team had itself recommended — leads under block-CV (0.395 against 0.255 for the bare core) and loses under both strict protocols (0.123 against 0.246 under Buffered LOO 300 m). It is therefore reported as **tested and rejected at this sample size**, not deferred to future work. The residual is trained at every run and `apply_residual` is false.

5.3 The delivered model

A line-source attenuation kernel with three positive parameters and no morphological input at all:

$$E = \frac{A_{\text{hw}}}{\max(d_{\text{hw}}, D_0)} + \frac{A_{\text{res}}}{\max(d_{\text{res}}, D_0)} + B, \quad L = 10 \log_{10} E,$$

with $A_{\text{hw}} = 4.774 \times 10^7$, $A_{\text{res}} = 3.800 \times 10^7$ and $D_0 = 5 \text{ m}$.

The background term is not identifiable from these data

B is fitted to -99.6 dB, a boundary solution. Called a three-parameter model, it behaves as a two-parameter one: 363 points over 3 sites do not constrain a floor.

Under the reference protocol: $R^2 = 0.246$ [0.097, 0.339], MAE = 5.01 dB, RMSE = 6.20 dB, against a data spread of 7.15 dB. **A typical prediction is wrong by about 5.01 dB** — a factor of three in acoustic energy. That is enough to support a contrast and nowhere near enough to support an assessment.

Note also that the confidence interval of the delivered model overlaps that of the plain $\log(d_{\text{road}})$ regression ($R^2 = 0.200$). The ordering between those two is not statistically separated at this sample size, and should not be reported as though it were.

5.4 Generalisation to an unseen typology

Under leave-one-site-out, the learned models collapse towards or below zero while the physical kernel retains most of its score. With three sampled typologies the number of independent morphological configurations available is close to three, whatever the number of points — so maps are clipped to the sampled envelope regardless of the score.

6 Agent-based simulation

The GAMA model (`simulation/gama/hanoi_noise.gaml`) reads the exported 40 m grid and adds receiver agents, mobile vehicles for visualisation, and construction sites calibrated in energy. Tier 1 exposes sliders for hour, traffic volume and mitigation.

One correction worth recording. The traffic multiplier scales the *traffic share* of the energy and never the background. Applied to total energy, it double-counts the background and inflates the apparent benefit of mitigation: the claimed gain from pedestrianisation fell from 7.0 dB to 3.5 dB once the multiplier was applied correctly.

Measured flow is injected only within 150 m of a measurement point. The exported zone extends 400 m beyond the survey envelope, and spreading flow uniformly over every street in it left the Hoan Kiem lake loop empty although every video had been filmed there.

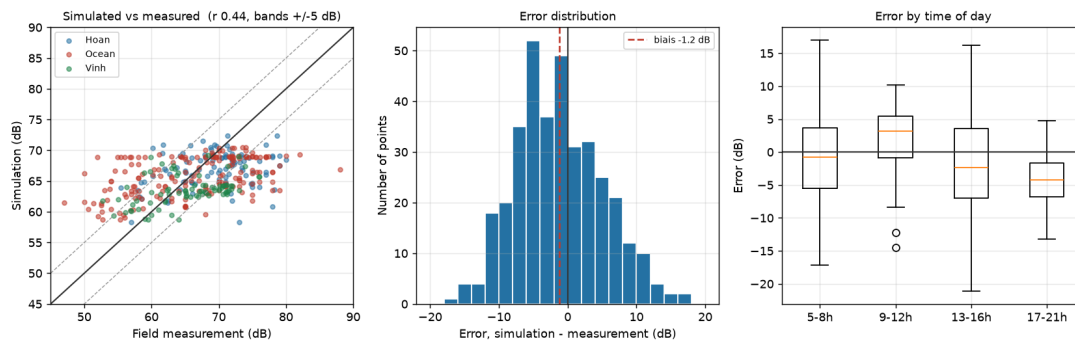


Figure 6: Simulated against measured levels. The simulation reproduces the spatial and temporal ordering; it does not reproduce the spread, which is expected given that features aggregated over 300 m cannot resolve the facade/courtyard contrast.

7 The field application

Every limitation above reduces to one fact: the campaign did not control its own instrument. The level came from a closed third-party application whose weighting, integration window and microphone gain were not inspectable; the form came from a second application; and pairing the two meant recording the level of one stretch of street and the audio of another. An Android application (`mobile/`, Kotlin and Jetpack Compose, `minSdk 26`) was built to replace that pairing and to carry the study.

7.1 Collecting

Both survey forms are transcribed as native screens, of 16 and 6 questions, and one microphone session produces both the level and the clip: 25 s A-weighted, SLOW, from `AudioRecord` after a discarded warm-up, with automatic gain control, noise suppression and echo cancellation switched off. The platform does not normally grant the microphone to two consumers at once, which is why measuring and recording were never separated into two steps. The GPS accuracy gate is the protocol's own, 10 m. Instances queue in an on-disk outbox and are submitted to KoboToolbox over OpenRosa when there is signal, so a survey round needs no network at all.

One integration detail worth recording, because it is invisible from the device and cost a day. KoboToolbox discards the `XLSForm`'s `id_string` on deployment and substitutes the asset's own identifier. A submission naming the spreadsheet's identifier is answered 404 for a form the server has never heard of, which from the phone is indistinguishable from a wrong address. The application reads `/formList` and submits under whatever the server calls the form.

7.2 Showing, and the two routes to the simulation

The application carries the exported 40 m grid, the 363 measurements, the delivered kernel and `metrics.json` itself; `syncStudyData` copies them out of the repository at build time rather than transcribing them, so re-running the pipeline and rebuilding moves the application with it. Predictions outside the sampled envelope are refused rather than answered, as §5 requires.

GAMA is an Eclipse desktop platform and does not run on Android. The application reaches the simulation two ways, and they are not interchangeable:

- **Tier 1, recomputed on the phone.** A traffic multiplier from $\times 0.2$ to $\times 3$ applied over the grid it already carries. No network, and it agrees with the model itself to within 0.2 dB across that range.
- **The model itself, over gama-server.** The screen drives a headless GAMA on the network and draws the model's own display, which buys everything the exported grid cannot hold — the mitigation scenarios, the construction sites, the fleet by hour. It needs a server that is running and reachable; the grid needs nothing.

The same correction, found a second time. The first on-device decomposition subtracted the kernel's additive constant — $10^{-99.6/10}$, a subtraction of nothing — from a cell energy of order 10^6 , so it scaled the total after all and returned exactly $10 \log_{10} k$. At a fifth of the traffic it gave -7.0 dB where the model gives -3.7 . That is precisely the error recorded under the simulation above, rebuilt independently on a different platform. Taking the residual to be the zone's own ambience — the fifth percentile of its levels — is what makes the two agree, and a unit test now asserts that a multiplier of one moves nothing. The application carries 36 JVM unit tests over 31 Kotlin source files (6 679 lines); the weighting curve is pinned to the IEC table, and the pure-Kotlin parts — the form logic, the instance format, the weighting, the sample byte order — are covered without an emulator.

7.3 What the application must not be claimed to do

- **Its level is a different quantity from the campaign's**, and it goes in its own field with the device model, the OS version and the audio source the platform actually granted. The two never share a column.
- **It is not calibrated.** No reading from it has been compared to a sound level meter; an emulator records silence. The microphone offset is a plausible constant, not a measurement, and the application says so on screen. Until that comparison is made the absolute figure means nothing — which, per §3.4, is equally true of the campaign's own numbers.
- **It makes no compliance claim.** The colour bands are the same QCVN references used descriptively elsewhere in this report.
- **It does not embed GAMA.** Drawing a model's display and driving it over a socket is not running it on the device.

Build, installation, and the two routes for submitting to KoboToolbox — named collectors, or the anonymous mode a publicly distributed APK requires — are in `mobile/README.md`; the internal design and the `gama-server` protocol are in `mobile/ARCHITECTURE.md`.

8 Limitations, stated plainly

- **The learned component does not survive an honest spatial split.** The hybrid architecture leads under the permissive 600 m block split and loses under both strict ones. The delivered model is the bare physical core; the residual is trained but not applied. Morphology aggregated over 300 m adds no measurable value over the two distance terms alone.
- **Absolute calibration.** The three phones are calibrated against each other, never against a reference instrument. A bias common to all three (+3.0 to +7.3 dB, bounded by anchoring on instrumented Vietnamese campaigns) is invisible in our data by construction. The field campaign is closed and this cannot be repaired retrospectively.

The question this invites is fair and should be answered rather than deflected: *was a single side-by-side reading against a class 1 or 2 meter genuinely out of reach?* A half-day rental would have settled the absolute scale, and it is by a wide margin the cheapest correction available to this work — one measurement turns every level here from calibrated in relative terms into calibrated. It was not done, and no reading in this report should be quoted as though it had been.

- **Measurement quantity.** A 20–30 s sample is not the L_{Aeq} of any regulatory reference period. A single horn burst moves it by several decibels — horn events reach +17 dB in Vietnamese traffic. This variance is a property of the quantity and caps the R^2 any spatial model can reach.
- **Generalisation — and this binds harder than the R^2 does.** The number of *independent morphological configurations* in this study is close to **three**, whatever the number of points: 363 measurements do not buy 363 measurements' worth of generalisation, because they are three draws from the space of urban fabrics and not 363 of them. The physical model extrapolates markedly better than the learned ones under leave-one-site-out, which is the column that measures this directly — but the limit is the sampling design, not the model. A fourth and fifth contrasting fabric would do more for the generalisation claim than any modelling change, which is why that item appears under next steps rather than as an aspiration.
- **Spatial resolution.** Features are aggregated over a 300 m radius, so adjacent 40 m cells share more than 98 % of their disc. The map cannot resolve the facade/courtyard contrast,

which reaches 10–15 dB in dense fabric: predicted levels are visibly flatter than measured ones.

- **Temporal coverage.** Only 10 measurements fall between 21:00 and 06:00, and none between 00:00 and 05:00 — although that is the period with the strictest threshold. One season only.
- **Traffic videos.** Counts are real flow, from tracking and line crossing, but speed remains unobservable without a ground homography and the field of view is not georeferenced, so distance is discarded. The detector was never validated against manual counts.
- **The field application is unvalidated where it matters most.** §7 describes a well-tested piece of software whose central number — the level itself — has never been checked against a sound source. It shares the campaign’s calibration limitation and adds its own: a second, uncharacterised microphone chain.

9 Next steps

- **Richer physics, not more learning.** The three-parameter line-source core already beats every learned variant. The next gain should come from a real propagation model — CNOSSOS-EU via NoiseModelling: screening, reflections, canyon geometry — not from more capacity fitted to 363 points. A learned residual should be re-tested only once the sample covers more typologies; on this one it degrades generalisation.
- **Anchor the absolute scale.** A class 1–2 sound level meter, or a 94 dB / 1 kHz acoustic calibrator, turns the bias interval into a measured correction. Everything downstream inherits it. This is now also the first task on the field application: calibrating it across levels and handsets turns its microphone offset from a plausible constant into an affine, measured correction, and it is the cheapest item on this list.
- **One instrument, more than one city.** With collection, level and submission in a single application, a second campaign can hold the instrument fixed — the same binary, the same window, the same gate. The Uganda transfer failure is over-determined precisely because the instrument, the protocol and the feature conventions all differed at once; deploying the same application twice separates them.
- **Traffic.** Flow by class is measured; *speed* is the missing half. Estimate it from track displacement under a ground homography, which also georeferences the field of view and so restores the source–receiver distance. Validate the detector against manual counts first.
- **More typologies.** A fourth and fifth contrasting urban fabric would do more for the generalisation claim than any modelling change.
- **Data publication.** Deposit measurements, forms and code with a DOI, and add an ethics statement covering the public-space video recordings.

10 Summary

363 smartphone measurements across 3 districts document a consistent exposure pattern in space and time. The delivered model is a three-parameter physical line-source law: $R^2 = 0.246$ [0.097, 0.339], MAE = 5.01 dB under Buffered LOO 300 m — ahead of a log(distance) regression (0.200), a six-feature LightGBM (0.137) and the physics-plus-ML hybrid built to improve on it (0.123).

Three negative results are the contribution. Every learned elaboration gains under a permissive spatial split and loses under a strict one. Cross-city transfer from Uganda fails. Video counts yield no emission coefficient, as density or as flow. **We claim contrasts, not absolutes.**

Generated by `scripts/10_build_report.py` from `data/processed/measurements.csv` and `models/metrics.json`.
Delivered model: `physical` — Physical kernel alone (3 parameters), selected under Buffered LOO 300 m. Reproduce
with `make report`.